

RESEARCH DOCUMENT

Literature Survey

Researcher and Author
Howida Gharib Saad El Din Selim

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Survey Methodology

The survey prioritizes 2021-2026 publications while retaining foundational Hamming, GDI, and FS-GDI sources. Publisher pages and DOI records are used to verify existence. Unavailable metadata are marked rather than inferred.

Comparative Survey

Reference	Year	Circuit	Logic	Contribution	Limitation
Hamming	1950	Error-correcting codes	Coding theory	Foundational Hamming-code theory	No circuit implementation
Morgenshtein Fish Wagner	2002	Combinational gates	GDI	Introduced GDI methodology	Reduced swing and nonstandard body/source concerns motivate restoration
Morgenshtein Yuzhaninov Kovshilovsky Fish	2014	16-bit CLA	FS-GDI	Full-swing GDI and robustness methodology	CLA rather than ECC encoder
Radhakrishnan Nirmalraj Karn	2019	Logic gates and full adder	Enhanced GDI	Enhanced-GDI restoration evaluated in LTspice	Different node and very different loads/frequencies
Hasan Zaman Hossain et al.	2020	Scalable 1-bit full adder	Hybrid full-swing GDI	Hybrid GDI full-adder implementation	Arithmetic cell rather than Hamming encoder
Publisher record: GDI synthesis study	2021	Combinational synthesis	GDI with MUX/BDD decomposition	Systematic mapping of Boolean networks	Not specific to ECC and full metadata needs subscription verification
Ghorbani Dolatshahi Zanjani Barekatain	2022	Full adder and XOR	Dynamic-GDI	Combines dynamic logic and GDI	Non-CMOS CNFET results are not directly comparable
Tripathi Jana Bhaumik	2023	SEC-DAEC and SEC-DAEC-TAEC codecs	HDL/synthesized architectures	Stronger adjacent-error correction codes	More complex codes and incompatible methodology
Bahrami Shiri Pesaran	2023	Approximate subtractors	GDI with dynamic threshold	Explores compact GDI structures	Approximate logic is unsuitable for exact ECC parity
Shiri et al.	2023	Approximate full adders	GDI/dynamic threshold	Energy-efficient GDI approximate adders	Approximation violates exact encoding requirements
El-Bendary El-Badry	2024	Hamming (7,4) and (11,7)	4) and (11,7)	72 for FS-GDI Hamming (11,7)	7)
Albert Thiagarajan Sinthia Christopher Suganthi Arunkumar	2024	Boolean functions and full adder	Optimized FS-GDI	Warns that direct FS-GDI mapping is function dependent	Journal indexing/metadata quality should be checked before final submission
IEEE ESCI conference paper	2024	Hamming encoder and decoder	GDI versus CMOS	Recent GDI Hamming codec application	Insufficient accessible setup for direct quantitative comparison
Kim et al.	2026	Long Hamming encoders/decoders	Word-serial architecture	Word-serial trade-off for long codes	Targets long codewords rather than transistor-level short encoders

Identified Gap

Foundational work established Hamming codes, GDI, and FS-GDI. Later work repeatedly demonstrated reduced transistor count or energy in arithmetic cells, and the baseline 2024 paper applied a six-transistor FS-GDI XOR to Hamming (7,4) and (11,7). Recent ECC research also covers SEC-DED and stronger adjacent-error codes. Therefore, neither extended Hamming nor GDI alone is novel.

The defensible gap is narrower: the accessible literature does not provide a reproducible, fair, transistor-level comparison of CMOS, reconstructed FS-GDI, and selectively restored FS-GDI/CMOS extended Hamming

(12,7) encoders using one predictive model, one activity pattern, explicit D1-to-P0 delay measurement, automated `.meas` extraction, and controlled VDD, temperature, load, and frequency sweeps.

The baseline paper stops at Hamming (11,7), reports no overall parity output, does not publish its netlists, and contains inconsistent delay-improvement percentages. It also treats transistor count as area. This thesis addresses those methodological weaknesses and tests two limited hybrid candidates rather than assuming that buffering improves every metric.

The resulting evidence shows a real trade-off. Hybrid-B is faster than CMOS and FS-GDI on the measured D1-to-P0 path, while FS-GDI retains the lowest simulated power and PDP. The scientific contribution is therefore a traceable characterization and a speed-oriented selective-restoration architecture, not a claim of universal superiority.